

INTERMEDIATE MICROECONOMICS
CONSUMER THEORY
(FOLLOWING VARIAN, CHAPTERS 2–5)

Aldermast & Cross Consider the Menu

*Being an account of budget constraints, the utility function,
indifference curves, the tangency condition, and an expense
form that will require careful itemisation*

FOREWORD

The University of Applied Consequence holds its annual conference dinner on the last Friday of November. It has done so since 1974. Professor Aldermast attended every one of them.

SCENE 1

The slope of the budget constraint is the rate at which the world allows you to exchange one good for another.

PROFESSOR ALDERMAST IS THE DEAN OF MARGINAL AFFAIRS at the University of Applied Consequence. He has held this position for eleven years. He wants wine. He has always wanted wine. This has never been a matter of uncertainty.

Professor Cross holds the Chair in Indifference. His research demonstrates that consumers maximise utility subject to a budget constraint. He wants food. He has wanted food since the morning seminar, which ran forty minutes over.

They have been colleagues for thirty-one years and have several areas of dispute between them. Cross had published eleven papers on the disputed questions. Aldermast had not read them. Both considered the arrangement satisfactory.

The department has allocated £200 for the evening. This is described in the faculty handbook as a generous allowance commensurate with the Institution's standing. Gerald has been their waiter for eleven years. Gerald has views.

Gerald placed the menus on the table and did not leave.

"Two hundred pounds," said Gerald. "For the evening. Between you. A course of food is twenty pounds. A glass of wine is five pounds."

"We are aware of the budget," said Aldermast.

"He is constructing the feasible set," said Cross.

"He is a waiter," said Aldermast.

"I prefer to think of myself," said Gerald, "as a price vector."

“You are a waiter,” said Aldermast.

“I am a waiter,” said Gerald, “who has been presenting these two prices at this table for eleven years.”

Cross took out a pen. He wrote on the back of the menu. Aldermast watched this with the expression of a man who has seen it happen many times and has not found a way to prevent it, having tried.

“Two goods,” said Cross. “Call food x_1 and wine x_2 . The price of food is $p_1 = 20$. The price of wine is $p_2 = 5$. Our total income for the evening is $m = 200$. We can spend our income on any combination of the two goods, provided our total expenditure does not exceed the budget and we consume non-negative quantities of each. We cannot donate food back to the kitchen. We cannot return wine. That gives the budget set:

$$B = \{(x_1, x_2) : p_1x_1 + p_2x_2 \leq m, x_1 \geq 0, x_2 \geq 0\}.$$

The upper boundary of this set — the budget line itself — is where expenditure exactly exhausts the budget:

$$p_1x_1 + p_2x_2 = m.$$

Every bundle (x_1, x_2) on this line costs exactly m . Every bundle strictly inside the set is affordable but wasteful. Every bundle outside it is not available to us.”

“Trust you to speak in abstractions,” said Aldermast. He took the pen and wrote: $20x_1 + 5x_2 = 200$. Thirty-one years of Cross had left a mark.

“Indeed,” said Cross, taking back the pen. “The feasible set is fixed.”

To understand the shape of the constraint, Cross rearranged it. Solving for x_2 :

$$x_2 = \frac{m}{p_2} - \frac{p_1}{p_2}x_1 = 40 - 4x_1.$$

This is a straight line. Its slope is $-p_1/p_2 = -4$. The slope is the rate at which the world — not Aldermast, not Cross, the world — allows them to exchange food for wine. Every additional course (x_1) costs four glasses of wine (x_2). Not because Cross prefers it that way. Because the à la carte menu dictates it.

The line has two intercepts. Setting $x_1 = 0$: they could have no food and forty glasses of wine, spending the entire £200 on wine alone. Setting $x_2 = 0$: they could have ten courses and no wine. These are the extremes. Every point in between is a feasible bundle.

“At one extreme,” said Cross, “ten courses. No wine.”

“Unacceptable,” said Aldermast.

“At the other extreme, no courses. Forty glasses of wine.”

“Now we are discussing something,” said Aldermast.

“Both extremes are on the budget line,” said Cross. “So is every combination in between. The slope of -4 tells us the trade-off the market imposes.

Consider what happens if something changes. If our income rises to $m' > m$, the constraint becomes $p_1x_1 + p_2x_2 = m'$.

The line is

$$x_2 = \frac{m'}{p_2} - \frac{p_1}{p_2}x_1.$$

The slope $-p_1/p_2$ is unchanged — the prices have not moved, so the rate of exchange between food and wine is the same. But both intercepts increase: we can now afford m'/p_1 courses at one extreme and m'/p_2 glasses at the other. The entire constraint shifts outward.

Now suppose instead that the price of food rises to $p'_1 > p_1$. We have

$$x_2 = \frac{m}{p_2} - \frac{p'_1}{p_2}x_1.$$

The wine intercept m/p_2 is unaffected — if we spend everything on wine, the price of food is irrelevant. But the food intercept falls from m/p_1 to m/p'_1 , since food now costs

more per course. The slope steepens from $-p_1/p_2$ to $-p'_1/p_2$. Food has become more expensive relative to wine. The constraint pivots inward on the wine intercept.”

“Neither of those things has happened,” said Aldermast. “We have two hundred pounds and the prices are as stated.”

“I am explaining the general case,” said Cross.

“The general case,” said Aldermast, “is not on the menu.”

Gerald refilled the bread basket.

SCENE 2

A preference relation is rational if and only if it is complete and transitive.

VIVA WAS AT THE ADJACENT TABLE. She had been at the adjacent table for twenty minutes. She had the menu open. She had closed it. She had opened it again. The menu had not changed. This had not helped.

She was in the third year of her doctorate at the University of Applied Consequence. Her supervisor had submitted a conference abstract on her behalf without telling her. The abstract was titled *Rational Agency: A Preliminary Assessment*. Viva had not written it. Her supervisor said this happened to everyone. Viva was not certain this was true but could not immediately refute it.

The problem, had she known to name it, was completeness. The first condition of a rational preference relation requires that for any two options x and y , the consumer can declare $x \succsim y$, or $y \succsim x$, or both (meaning indifference). The strict preference relation $x \succ y$ holds when $x \succsim y$ but not $y \succsim x$ (x is preferred to y , and y is not preferred back). Viva could, for the moment, do none of these things. Fish against lamb: no ranking. Lamb against vegetarian: no ranking. Fish against vegetarian: no ranking. She had three options and could not rank any pair of them. Without completeness, the preference relation does not exist. There is nothing to represent, nothing to maximise, nothing to work with. Nothing to order.

Gerald arrived at her table.

“I am uncertain,” said Viva.

Gerald did not write that down. He waited.

“I much prefer the fish to the lamb,” said Viva. “And the lamb to the vegetarian. But given a choice between fish and vegetarian, I’d rather have the vegetarian.” She stopped. “No.

The fish.”

“Of course,” said Gerald.

Cross had been listening. He turned to Aldermast. “Two problems,” he said. “First she could not rank the options at all. That is completeness failing — no preference declared over any pair. Then she produced an irrational ranking. Strictly fish over lamb, strictly lamb over vegetarian, strictly vegetarian over fish. So she strictly prefers fish to fish. That is transitivity failing — the ranking cycles. Then she caught it. Corrected it. The preference relation exists.”

Completeness was the first condition. Transitivity is the second. It requires that if $x \succsim y$ and $y \succsim z$, then $x \succsim z$. Rankings must be coherent across triples. If fish is preferred to lamb and lamb to vegetarian, then fish must be preferred to vegetarian. A ranking that cycles — as Viva’s initially did — violates this. It is not merely inconvenient. It means no utility function can be constructed. To assign numbers compatible with her initial ranking, we would need:

$$u(\text{fish}) > u(\text{lamb}) > u(\text{veg}) > u(\text{fish}).$$

A number cannot be strictly greater than itself.

“The utility function cannot exist,” said Aldermast.

“Correct,” said Cross.

“No ranking. Then a cycle. Then a correction,” said Aldermast.

Gerald returned to their table. “She always orders the fish,” he said. “She has been coming to this conference for three years.”

“The axioms characterise the preference relation,” said Cross. “Not the difficulty of articulating it. The axioms are the conditions that preferences must satisfy to be qualified as rational.”

“Whether any given diner meets those conditions,” said Gerald.

“Is an empirical matter,” said Aldermast.

SCENE 3

A utility function exists if the preference relation is complete, transitive, and continuous. It may not be unique.

CROSS HAD NOT PUT DOWN THE PEN. “We established that preferences must be complete and transitive,” he said. “The question is what those conditions buy. And what they do not.”

“I have not ordered,” said Aldermast.

What they buy is existence. Provided a preference relation is complete, transitive, and satisfies a third condition — continuity — a utility function can be constructed. The precise claim is this: there exists a function $u : \mathbb{R}_+^2 \rightarrow \mathbb{R}$ such that for any two bundles (x_1, x_2) and (y_1, y_2) ,

$$(x_1, x_2) \succsim (y_1, y_2) \iff u(x_1, x_2) \geq u(y_1, y_2).$$

The preference relation and the utility function carry identical information. The function assigns a real number to every bundle. The number encodes its rank. This is exactly what the axioms make possible: completeness ensures every bundle receives a number, because the consumer can rank every pair; transitivity ensures the numbers arrange themselves into a coherent order, because the ranking does not cycle. Remove completeness and some bundles cannot be ranked — the function is undefined on those pairs. Remove transitivity and the required order contradicts itself — as it would have for Viva, had her preferences genuinely cycled. To assign numbers consistent with fish over lamb, lamb over vegetarian, and vegetarian over fish, one would need a number simultaneously greater than, and less than, itself. The assignment fails. The utility function cannot exist.

“They are the minimum conditions under which a utility function can be written down at all,” said Cross. “But not the only ones. Continuity is what completeness and transitivity

do not give you. Consider what happens without it. Suppose Aldermast's preferences over wine were lexicographic — any bundle with more wine strictly preferred to any bundle with less wine, regardless of food —”

“Lexicographic,” said Gerald. “Like a dictionary. You don't consult the second letter until the first letters match.”

“Wine first,” said Aldermast. “Always.”

“Always,” said Cross. “Now fix the wine level. Say two glasses. At exactly two glasses of wine, every possible quantity of food is strictly ranked — one course is preferred to none, two to one, and so on without limit. To represent that ranking with numbers, you need an entire number line. One for every food quantity at that wine level alone.”

“One number line,” said Aldermast.

“One number line,” said Cross. “Now move to three glasses. Everything with three glasses of wine is strictly preferred to everything with two, regardless of food. So the utilities at three glasses must all sit strictly above the utilities at two glasses. Another entire number line, stacked above the first.”

“And there are infinitely many wine levels,” said Aldermast.

“Infinitely many,” said Cross. “Each requiring its own number line, strictly above the last. One number line is not enough. No utility function can exist.”

“I do not have lexicographic preferences over wine,” said Aldermast.

“They would be perfectly rational if you did,” said Cross. “Complete and transitive. Simply not representable by a utility function. The number line would not be sufficient.”

Aldermast considered this. He was looking at the wine list.

“A glass of Burgundy is better than no glass,” said Aldermast. “One could give up an infinity of food.”

“That,” said Cross, “is what continuity rules out.”

“It rules out the utility function,” said Aldermast. “The preference remains rational.”

“Correct,” said Cross.

Aldermast put down the wine list. He would not have himself accused of lexicography.

Cross wrote on the menu again. Next to $u(x_1, x_2)$ he wrote $v = 2u$, and below it $w = \ln u$. Three functions. He pushed the menu toward Aldermast.

“The same preferences,” said Cross. “Any strictly increasing transformation of a utility function represents exactly the same preference relation. The numbers change. The ranking does not. The choice of utility function is, in this sense, arbitrary.”

“So the numbers are meaningless,” said Aldermast.

“Ordinal,” said Cross. “They rank correctly. They do not measure anything beyond the rank. It is an existence result, not a uniqueness result. The function exists. Which function is arbitrary.”

“So when I tell people that the venison is twice as good as the lamb,” said Gerald, “I am making a claim the preference relation cannot support.”

“You are,” said Cross.

“I have been doing that for eleven years.”

“The preference relation was correctly communicated,” said Cross. “The magnitude was unwarranted.”

“Reassuring,” said Gerald, and left.

Cross looked at the menu. Then, for a moment, at Aldermast. Aldermast was looking at the wine list. He had lost weight since October. Cross looked back at his pen.

SCENE 4

The slope of the indifference curve is the rate at which you, personally, are willing to exchange one good for another.

*** DRAFT SCENE - NEEDS TO BE CHECKED ***

THE MENU HAD SUFFERED. Cross had covered the back of it in a notation that accumulated over the course of the evening: the budget constraint, its intercepts still legible; the utility functions, stacked carefully: $u(x_1, x_2)$, $v = 2u$, $w = \ln u$. The front of the menu listed five dishes and a selection of drinks. The back listed, with greater precision, the framework under which any of them might be chosen.

Cross uncapped his pen.

“You are not lexicographic,” he said.

“I said that.”

“Then at some bundle of food and wine you are indifferent between that bundle and a neighbouring one. The rate at which you would trade, at that point, remaining indifferent — that rate exists. It is yours specifically.”

“I have not agreed to a rate.”

“You agreed to continuity. The rate follows. And the set of all bundles between which you are precisely indifferent forms a curve.”

Aldermast looked at him. “I am not a curve.”

“You have preferences. Your preferences have a structure. That structure, when represented geometrically, is a curve. Several curves, in fact. One for each level of utility.”

“Then I have several curves.”

“You do.”

“And what do they look like.”

Cross drew.

He found a clear margin on the menu — a narrow one, near the wine list — and drew two axes. Food along the horizontal. Wine along the vertical. He drew a curve, convex to the origin, smooth, bowing inward toward the lower left. Then he drew another, farther from the origin. Then a third.

“These are your indifference curves,” he said. “Every point on a given curve is a bundle between which you are precisely indifferent. Every point on a higher curve is preferred to every point on a lower one.”

“The higher ones are better.”

“More wine, more food. Yes.”

Aldermast regarded the curves with the expression of a man who had been correctly described and did not enjoy it. “And the shape.”

“Convex to the origin. This reflects diminishing marginal rate of substitution. The more wine you have, the less you are willing to give up food to acquire another glass. The more food you have, the less it is worth to you — and so the more of it you would trade for wine.”

“The more food I have,” said Aldermast, “the less I want more of it relative to wine.”

“Precisely.”

“That is true of most things.”

“It is the economic formulation of it, yes.”

Viva appeared at the edge of the table.

She had come from the adjacent table, where her fish had been sitting for some minutes. She stood at the edge — not pulling out a chair, not intending to — with the look of someone who had been waiting for a pause that had not come.

“The chapter three result, Sir,” she said. “On the existence conditions. I wanted to ask—”

“The monotonicity assumption carries it,” said Aldermast. He had not looked up from the menu. “It is in the footnote.”

A beat.

“Yes,” said Viva. “I know. I meant the continuity—”

“The footnote,” said Aldermast, and turned back to Cross.

She stood there a moment longer. Then she did not leave — she had not finished, or was not ready to be dismissed, or both.

Aldermast picked up his water glass. A droplet ran down the outside. It fell.

It landed on the menu at the exact point where the first and second curves approached each other most closely. The ink blurred. The two curves, in the spreading wet, appeared to merge — and then, as the droplet spread further, to cross.

Cross looked down. He went still.

Two indifference curves. Crossing.

“That cannot happen,” he said.

“The water fell.”

“The curves cannot cross.” He said it quietly. “If they cross, there is a point on both curves simultaneously. A bundle you are indifferent to on curve one and also indifferent to on curve two. But a bundle on curve two is strictly preferred to a bundle on curve one. You cannot simultaneously strictly prefer and be indifferent to the same bundle. It is a contradiction. It violates transitivity.”

“I didn’t draw them crossing.”

“The water drew them crossing.”

“Then the water is irrational.”

Viva looked at the smudge. She reached across.

She had a pencil. She set the edge of it against the menu and drew a short, clean arc — correcting the apparent intersection, pulling the lines apart, restoring the gap between them.

Then she went back to her plate.

Cross looked at the correction. Then at Viva. Then at the corrected curves, which were once again properly ordered — separated, non-crossing, each higher curve strictly preferred to the lower.

He looked at Aldermast.

Aldermast kept his gaze on the smudge.

“And the slope,” he said.

Cross recapped his pen. He uncapped it again. “The slope of an indifference curve at a given point is the marginal rate of substitution. The MRS. It measures the rate at which you are personally willing to trade food for wine, at that particular bundle, remaining indifferent.”

“My rate.”

“Yours specifically.”

Aldermast considered this. “The slope is subjective.”

“The slope is your preferences. Your preferences determine the shape of the curves. The shape of the curves determines the slope. The MRS is not a market rate. It is not what the world will let you exchange. It is what you, personally, are willing to exchange — from the inside.”

A brief silence.

“Write it down,” said Aldermast.

Cross wrote. He found another margin near the printed heading *La Conséquence*.

$$\text{MRS} = \left. \frac{dx_2}{dx_1} \right|_{u=\bar{u}} = -\frac{MU_1}{MU_2}$$

“The ratio of marginal utilities,” said Cross. “How much you gain from another unit of food, relative to how much you gain from another unit of wine. That ratio is the slope of the curve — how much wine you would surrender for another course.”

“And the specific form.”

“I was coming to that.”

Cross moved the pen to a new margin. There were not many left. The menu was, at this point, largely mathematics.

$$u(x_1, x_2) = x_1^\alpha x_2^\beta$$

“Cobb-Douglas,” said Cross. “A particular functional form for the utility function. Two parameters — alpha and beta. Your preferences, in this form, are characterised by these exponents.”

“And what do alpha and beta mean.”

“The ratio of alpha to beta determines how much of your income you spend on food relative to wine. If alpha is one and beta is three — as I would estimate, given your behaviour over eleven years of dinners — you spend one quarter of your income on food and three quarters on wine. Fixed. Regardless of price.”

“Fixed.”

“The Cobb-Douglas utility function generates constant expenditure shares. Whatever the price of wine, whatever the price of food, you always allocate three quarters of your budget to wine.”

Aldermast looked at the equation. He looked at the ratio. He looked at Cross.

“And you have estimated my alpha and beta from observation.”

“Eleven years of dinners.”

“And you consider this permissible.”

“I consider it,” said Cross, “accurate.”

A pause. Aldermast looked at the equation a moment longer. Then he looked at his empty wine glass.

Across the room, Gerald appeared.

He had appeared at regular intervals throughout the evening. Each time the question had been the same. He stood for a moment at the edge of the table — not approaching, not withdrawing — in the manner of a man who has been professionally patient for eleven years and is finding the evening unusually long.

Aldermast looked up. Gerald looked at the menu. At the mathematics on the menu. At Cross's pen. At Aldermast.

"The MRS has been established," said Cross. "We are very nearly ready."

"The kitchen—" said Gerald.

"Very nearly."

Gerald withdrew.

SCENE 5

Last scene goes here

*** DRAFT SCENE - NEEDS TO BE DRAFTED ***

EPILOGUE

The expense form will require careful itemisation.

GERALD BROUGHT THE FORM. It was a standard departmental form. Two columns. Two goods. Quantities and prices. There was a line for the Dean's countersignature at the bottom.

Cross filled it in. Food: zero courses for Aldermast, zero for himself. Wine: zero glasses for both. Total expenditure: zero pounds. The department had allocated two hundred. The bread basket had been complimentary. Gerald had refilled it once.

Cross slid the form across the table.

Aldermast looked at it for some time. Two columns. Four zeros. Two hundred pounds unspent. An entire evening accounted for in the space of four numbers, all of them the same number.

He looked at Cross.

Cross had his pen ready.

Aldermast took the form. He turned it over. He wrote something on the back. Not a signature. Something else. Cross leaned slightly. Gerald moved the form before he could see it.

Viva passed the table on her way out. She had her notebook under her arm. She paused. She looked at the two of them — at the pen in Cross's hand, at the menu on the table, at the form in Gerald's. She said nothing. She left.

Cross picked up the pen. He held it for a moment. Then he put it down.

"The model was misspecified," said Cross.

Aldermast said nothing.

“Food and wine,” said Cross. “Two columns. That is what the form asks for. That is what the model optimises over. It does not have a column for thirty-one years. It cannot represent the value of a colleague who has not read your papers and therefore cannot tell you they are wrong. It has no price for this argument, or the one before it, or the arrangement that has been satisfactory to both of us since 1997. These are not goods the market provides. There is no budget line for them. No intercept. No slope.”

Aldermast said nothing for a moment.

“Eleven papers,” he said.

“Yes,” said Cross.

Gerald took the form. He folded it. He placed it in his breast pocket rather than on his tray.

He had never done that before.

“Will there be anything else,” he said.

Aldermast looked at the wine list. He closed it.

“Three glasses,” said Aldermast. “One for you, Gerald. And bring me the bill separately.”

Gerald did not write it down. He already knew.